1. Introduction

The objective of this project is to classify building types including single_family, apartment_condo, commercial, mixed_use ,empty_land, unknown using Google Street View images . The classification of building types from street-view imagery is a challenging computer vision task that combines elements of structural recognition, contextual analysis, and functional inference. Unlike traditional image datasets where objects are centrally framed and clearly labelled, street-view images present several complications. Buildings are often partially visible due to occlusions from trees, vehicles, or other urban infrastructure. Lighting conditions vary across times of day and seasons, and images are captured from multiple viewpoints that can distort perception. As a result, identifying building types such as single-family homes, apartment or condo complexes, and commercial properties requires careful interpretation of subtle visual and contextual cues rather than explicit indicators.

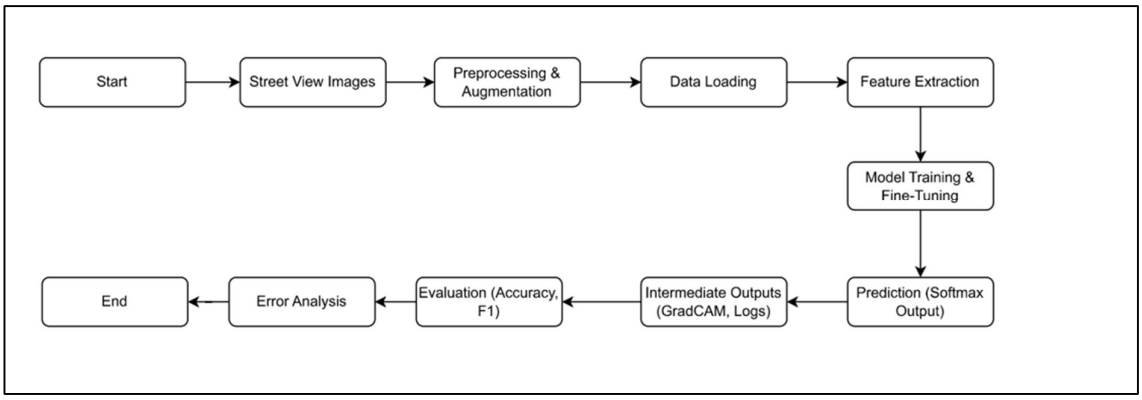
Street-view imagery introduces ambiguity that complicates classification. For instance, multi-story residential buildings and small apartment complexes may share architectural elements, such as repeated window patterns or balconies. Commercial buildings can resemble large apartment complexes if signage is obscured. For the current Dataset we have a clear class imbalance: residential buildings are represented, while commercial properties are underrepresented in both training and test sets. Additionally, the dataset contains only 104 training samples and 88 test images, which limits statistical power and increases the risk of overfitting drastically. These characteristics necessitate careful pipeline design and evaluation metrics that account for class imbalance and visual ambiguity.

2. Visual Cues and Data Characteristics

Accurate classification relies on three primary types of cues: structural, functional, and contextual. Structural cues include the number of floors, building height, facade symmetry, window and balcony patterns, and roof characteristics. For example, the location 1341 Lasalle St, Charlotte, NC, displays multiple entries to the apartment and repeated window patterns indicative of an apartment/condo structure. Functional cues relate to building usage, such as the number of entrances or visible commercial signage. Single-family homes typically exhibit a single entrance and private driveway, whereas apartment properties often include multiple entrances and visible storefronts. Contextual cues involve the surrounding environment, such as no of cars, street density. A large parking lot with minimal landscaping may suggest a commercial building, while private lawns and driveways are indicative of residential use.

Dataset limitations introduce challenges in interpreting these cues. Many residential subclasses, particularly single-family homes and small apartments, share overlapping features, making classification uncertain. Additionally, no data of commercial buildings in the dataset reduces model exposure to varied patterns, increasing the likelihood of misclassification. These constraints highlight the importance of multi-faceted feature extraction and careful evaluation.

3. Pipeline Design



The classification pipeline consists of several interdependent stages, each addressing a specific aspect of the task. The first stage, data preprocessing, standardizes input images to 224×224 pixels and normalizes pixel values to align with pretrained model requirements. Given the limited dataset size, data augmentation is critical. Techniques such as random cropping, horizontal flipping, rotation, and color jittering artificially expand the dataset and reduce overfitting, ensuring models generalize to unseen images.

The core of the pipeline, feature extraction and model training, leverages transfer learning on five architectures: VGG16, ResNet50, EfficientNet-B0, MobileNet-V2, and Vision Transformer (ViT-B/16). VGG16 was chosen for its balance of depth and stability on small datasets, while ResNet50's residual connections allow for deeper networks without vanishing gradient issues. EfficientNet-B0 balances computational efficiency with accuracy, and MobileNet-V2 provides a lightweight solution suitable for deployment. ViT-B/16 replaces convolutions with attention mechanisms to capture global image relationships. All models utilize pretrained ImageNet weights, with early layers frozen to preserve general visual features and final layers fine-tuned for building type classification.

Predictions are generated through a SoftMax classification layer, producing class probabilities, confidence scores, and top-N predictions. GradCAM visualizations provide interpretability by highlighting image regions influencing the decision. These intermediate outputs, saved in JSONL and CSV formats, support debugging and allow inspection of model attention patterns .The pipeline is executed via the jupyter notebook but can also be ran using runpipeline.py, which automates preprocessing, training, evaluation, and visualization in a reproducible workflow.

4. Implementation and Code Organization

The implementation of the pipeline is primarily contained within a single Jupyter Notebook, supported by a lightweight script (runpipeline.py) used to execute the notebook end-to-end. Instead of a fully modular multi-file architecture, the system is organized in a sequential, cell-based structure where each stage of the pipeline is implemented step-by-step. This includes configuration setup, data preprocessing, model loading, training, evaluation, and output generation, all within the same environment.

The pipeline begins with a configuration section where paths, hyperparameters, and model selections are defined. Data preprocessing and augmentation are implemented using torchvision.transforms, ensuring that all images are resized to 224×224 and normalized consistently across models. A custom data loading mechanism using PyTorch's dataset and dataloader utilities is then used to handle batching, shuffling, and efficient data access. Model initialization is performed within the notebook by loading pretrained architectures such as VGG16, ResNet50, EfficientNet-B0, MobileNet-V2, and ViT-

B/16, with transfer learning applied by freezing early layers and fine-tuning the final classification layers.

Training and inference are also handled within the same notebook, where forward passes, loss computation using cross-entropy, and optimization are executed iteratively. Evaluation metrics such as accuracy, macro F1-score, precision, and recall are computed using scikit-learn utilities. Intermediate outputs, including prediction logs in CSV/JSONL format and GradCAM visualizations, are generated and saved to a results directory for further analysis. GradCAM integration is implemented directly within the notebook using the pytorch-grad-cam library to provide interpretability of model decisions.

One of the main implementation challenges was ensuring consistency across different model architectures, particularly in terms of input dimensions and preprocessing requirements. Early issues such as dimension mismatches and inconsistent normalization were resolved by standardizing all inputs to 224×224 and applying uniform transformations. Additionally, care was taken to handle file paths and missing data robustly using environment variables and dynamic path resolution. While the current structure is not fully modular, it is flexible enough to support experimentation with different models, augmentation strategies, and evaluation settings within a single, reproducible pipeline.

5. Experiments and Evaluation

The system was evaluated on 88 test images across 10 locations. Metrics included accuracy, macro F1-score, precision, recall, and top N accuracy. Given the dataset's class imbalance, macro F1 was prioritized to ensure underrepresented classes contributed equally to evaluation.

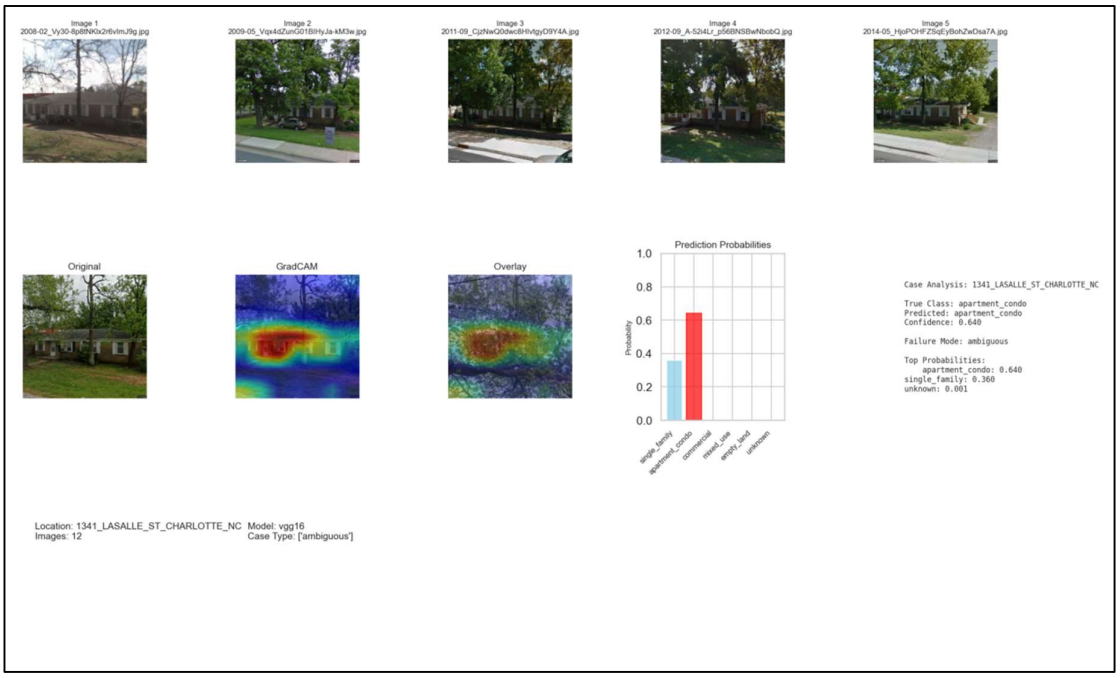
Model	Accuracy	F1-Macro	Precision-Macro	Recall-Macro	Top-2-Acc	Top-3-Acc
VGG16	1.0	0.5000	0.5000	0.5000	1.0	1.0
ViT-B/16	1.0	0.5000	0.5000	0.5000	1.0	1.0
EfficientNet-B0	0.875	0.4524	0.4583	0.4583	1.0	1.0
ResNet50	0.75	0.4028	0.4333	0.4167	1.0	1.0
MobileNet-V2	0.75	0.3833	0.4444	0.3889	1.0	1.0

VGG16 emerged as the top performer, suggesting that simpler architectures better mitigate overfitting on small datasets. Transfer learning enabled strong baseline performance despite limited data, with GradCAM visualizations confirming focus on meaningful building features such as facades, entrances, and balconies.

6. Error Analysis

A detailed error analysis was conducted to understand the limitations of the classification system by examining challenging and borderline cases. One representative case is the location 1341 Lasalle St, Charlotte, NC, where the model predicted the correct class (apartment/condo) but with relatively low confidence (~0.64–0.67). The difficulty in this case arises from structural ambiguity, as the building exhibits features common to both small apartment complexes and large single-family homes, such as limited balconies and a relatively compact structure. Additionally, partial occlusion and a restricted viewing angle reduce the visibility of distinguishing features like multiple entrances or shared amenities. GradCAM visualizations show that the model focuses on facade patterns and window alignment, but the lack of strong, distinctive cues limits confidence. This suggests that incorporating

multi-view imagery or contextual metadata, such as property type records or zoning information, could improve classification reliability in such ambiguous scenarios.



A second challenging case is observed at 1315 Beatties Ford Rd, Charlotte, NC, particularly in the controlled variation experiment. Under the baseline pipeline with augmentation, the model correctly classified the building as a single-family home with high confidence. However, when augmentation was removed, the model misclassified it as an apartment/condo with high confidence (~0.95–0.98). This indicates that the prediction was sensitive to training data diversity and that the model relied on superficial visual patterns, such as facade symmetry or nearby objects like parked vehicles, rather than true structural features. The misleading visual information in this case stems from limited variation in the training data, causing the model to overfit. GradCAM analysis further confirms that attention shifts to less relevant regions when augmentation is absent. This issue can be mitigated by maintaining strong augmentation strategies, increasing dataset diversity, and incorporating regularization techniques to improve generalization.

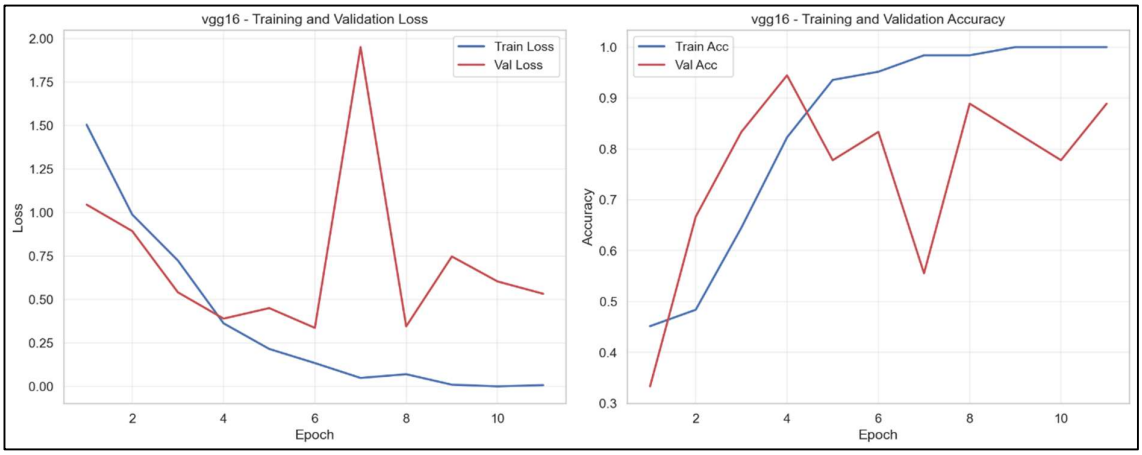
A third type of challenging scenario involves cases where useful visual information is largely absent or insufficient. In some street-view images, buildings are partially obscured by trees, vehicles, or shadows, making it difficult to identify key structural or functional cues. In such cases, the model may rely on weak contextual signals, such as surrounding environment or road structure, which can be misleading. For example, a residential building located near a larger road or open parking space may be incorrectly interpreted as commercial due to contextual bias. These situations highlight a fundamental limitation of relying solely on single-view imagery. To address this, the system could be improved by aggregating multiple images from different angles, incorporating temporal data, or using auxiliary data sources to provide additional context.

These cases collectively highlight the broader challenges of identifying building types from street-view images. One of the main challenges is the inherent ambiguity in visual features, where different building types share overlapping architectural characteristics. Additionally, variations in lighting, occlusion, and camera perspective further complicate feature extraction and interpretation. Another significant challenge is dataset imbalance, where certain classes, particularly commercial buildings, are underrepresented, leading to biased learning and reduced generalization. These factors make it difficult for the model to consistently distinguish between classes using visual information alone

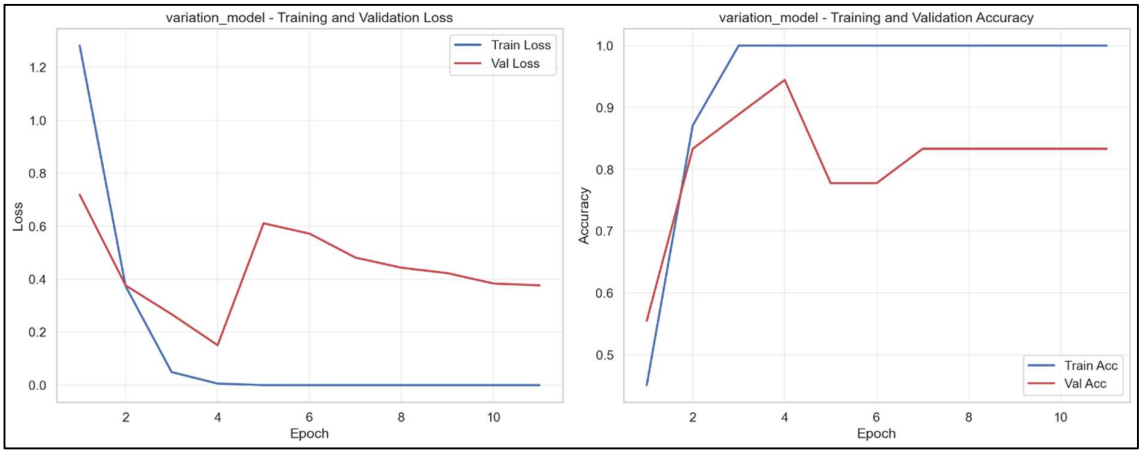
If additional data sources were available, several enhancements could be made to improve the system. Geographic metadata, such as zoning information or property records, could provide strong signals about building function. Multi-view street imagery or satellite images could offer complementary perspectives, helping resolve ambiguities caused by occlusion or limited viewpoints. Furthermore, incorporating temporal data or higher-resolution imagery could improve the visibility of structural details. Combining visual data with structured information would enable a more robust and context-aware classification system.

Scaling the system to handle 100,000 locations or an entire city would require significant redesign of the implementation. The current notebook-based pipeline would need to be replaced with a modular, production-ready architecture using distributed data processing and efficient storage systems. Data ingestion would need to be automated, potentially using APIs to collect street-view images at scale. Model inference would be optimized using batch processing and possibly GPU acceleration to handle large volumes of data efficiently. Additionally, results would need to be stored in databases, and monitoring systems would be required to track performance and detect failures. A scalable system would also benefit from cloud-based deployment and parallel processing frameworks to ensure efficient and reliable operation at scale.

7. Controlled Variation Experiment



VGG 16 Output with Augmentation



VGG 16 Output without Augmentation

The controlled variation experiment demonstrated the critical role of data augmentation in maintaining model generalization and stability. By removing augmentation while keeping all other parameters constant, the model exhibited a clear drop in performance, with macro F1-score decreasing from 0.5000 to 0.4028 and accuracy dropping from 1.0 to 0.75. Interestingly, prediction confidence increased in the absence of augmentation, indicating overfitting, where the model becomes overly certain about patterns that do not generalize well. The agreement rate of 75% between the baseline and variation further highlights that a significant portion of predictions depends on training data diversity.

A key example illustrating this effect is the Beatties Ford Road location, where the model correctly identified a single-family home under the augmented pipeline but misclassified it as an apartment/condo with high confidence when augmentation was removed. This shift was also reflected in GradCAM visualizations, where the model's attention moved from meaningful structural features to less relevant regions.. Overall, the experiment confirms that augmentation is essential for reducing overfitting, stabilizing feature learning, and ensuring robust performance, especially in small and imbalanced datasets.

8. Recommendations and Future Work

To improve system performance, several strategies are recommended: expanding dataset size and geographic diversity, capturing multi-angle images per location, balancing underrepresented classes, and incorporating structured metadata such as property age, zoning, and neighbourhood features. Ensemble models combining VGG16, ViT-B/16, and EfficientNet-B0 could reduce individual model biases. Multi-view aggregation, confidence thresholding, and attention-based weighting are suggested to handle ambiguous predictions. For large-scale deployment, distributed training, optimized data pipelines, and REST API model serving are advised. Multi-modal inputs, including imagery and structured data, could further enhance accuracy.

9. Conclusion

This research demonstrates a comprehensive pipeline for building type classification from street-view imagery. The system achieved a macro F1-score of 0.5000 with VGG16, illustrating the effectiveness of transfer learning and data augmentation on small, imbalanced datasets. GradCAM visualizations provided interpretability, error analysis identified failure modes, and controlled variation experiments highlighted the importance of augmentation. Future work will focus on dataset expansion, ensemble modelling, multi-view integration, and scalable deployment. The combined findings establish a solid foundation for robust, interpretable, and reproducible building classification in real-world applications.

10.AI Usage Disclosure

AI tools, like LLM such ChatGPT and code assistants like GitHub copilot, were used as a supporting aid for code implementation, debugging, and brainstorming pipeline improvements such as GradCAM. They also helped refine the clarity and structure of the report. All key decisions, including model selection, pipeline design, and analysis, were made independently, with AI mainly assisting in improving efficiency and presentation.

Reference:

Murdoch, R. and Al-Habashna, A.A., 2024. Residential building type classification from street-view imagery with convolutional neural networks. *Signal, Image and Video Processing*, *18*(2), pp.1949-1958.